

OPTIMIZATION OF GPS QUEUING DELAYS IN URBAN TRANSPORT LTE NETWORKS USING QUALITY OF SERVICE (QoS)

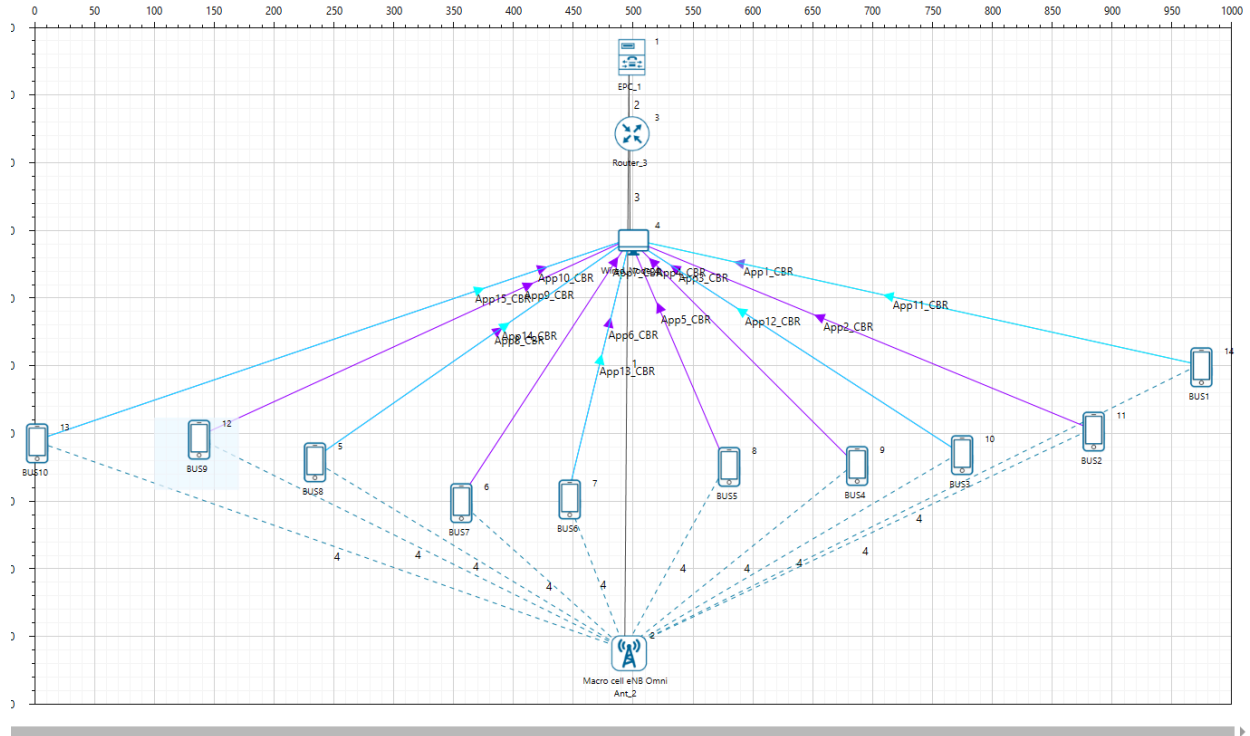
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SUMMARY

This project addresses the mitigation of critical latency in the transmission of geolocation data (GPS/RTPS) from an urban bus fleet operating in a high-congestion LTE network environment. The primary objective was to demonstrate the necessity and effectiveness of implementing Quality of Service (QoS) mechanisms to ensure that queuing delay does not violate the real-time Service Level Agreement (SLA).

Through comparative simulations of FIFO, Priority Queuing (PQ), and Weighted Fair Queuing (WFQ) policies, the project proved that the default policy (FIFO) introduces a critical delay of 760 ms. Adopting QoS (PQ/WFQ) completely solved the issue, reducing latency to 1.5 ms (a 99.8% improvement).

The conclusion identifies WFQ as the optimal solution, as it guarantees the necessary priority for GPS traffic without excessively penalizing low-priority traffic. A crucial technical clarification is included, acknowledging that the primary origin of congestion and delay in LTE occurs at the eNodeB's MAC Layer Scheduler (Layer 2), making the QoS applied at the Router (Layer 3) a valid demonstration of traffic classification effectiveness.



1. INTRODUCTION AND PROBLEM CONTEXT

1.1. Scenario and Relevance

Smart cities rely on the accuracy and timeliness of fleet tracking data. Bus tracking systems provide Estimated Time of Arrival (ETA) information, which must be accurate to maintain user trust and operational efficiency. In the simulated environment, the network infrastructure is based on LTE technology, chosen for its Wide Area Network (WAN) coverage and mobility management capabilities.

1.2. The Problem of Congestion and Delay

The problem arises when passenger traffic (large file downloads, streaming, labeled as Best Effort or BE) shares the same uplink path with critical GPS traffic (small, periodic packets, labeled as RTPS - Real-Time Polling Service).

During periods of high demand:

- **Head-of-Line Blocking (HoLB):** Large BE packets monopolize the router's queue.
- **Consequence:** Small GPS packets get stuck in the queue, resulting in excessive delay

that renders the location information obsolete before it reaches the server.

2. THEORETICAL FRAMEWORK AND TOPOLOGY

2.1. Network Architecture and Nodes

The simulation was implemented in NetSim using an LTE-Advanced network topology:

Device	Acronym	Function	Rolein Congestion
Buses	UEs (10 Nodes)	Source of all traffic (RTPS and BE) and mobile nodes.	Generate network saturation.
Tower	eNodeB	Wireless access point. Manages resource allocation (MAC Layer).	Origin of delay (Layer 2).
Core	EPC	LTE Core Network.	Backhaul network routing.
Router	Router_3	QoS Control Point. Simulates the backhaul bottleneck.	Where the queuing experiment is implemented.
Server	Wired Node	Final Destination.	Point of latency measurement.

2.2. Traffic Protocols and QoS Classification

Traffic was classified at the transport layer to enable differentiation:

- **RTPS (GPS):** Highest priority. Small packets (150 Bytes). Uses UDP due to its real-time, connectionless nature.

- **BE (Congestion):** Lowest priority. Large packets (1024 Bytes).

The simulation focused on the performance contrast between the extremes: RTPS vs. BE.

3. SIMULATION METHODOLOGY AND QUEUING STRATEGIES

3.1. Traffic Design and Mobility

- **Critical Load:** 10 RTPS flows, 1 packet/s.
- **Saturation Load:** 5 BE flows, configured at a rate that deliberately exceeds the Router's output link capacity to guarantee congestion.
- **Mobility:** Random Waypoint Model (velocity 15 m/s), simulating bus movement and stops.

3.2. Queuing Experiments

Three queuing disciplines were compared on the Router:

1. **Experiment 1: FIFO (Baseline):** Demonstrates the failure scenario due to Head-of-Line Blocking.
2. **Experiment 2: Priority Queuing (PQ):** Tests the strict priority solution for RTPS traffic.
3. **Experiment 3: Weighted Fair Queuing (WFQ):** Tests the fair sharing solution, assigning a weighted bandwidth quota to RTPS.

4. CRITICAL TECHNICAL ANALYSIS (Professor Feedback Integration)

The Layer 3 QoS solution is validated against the fundamental constraint of the LTE architecture.

4.1. Origin of the Bottleneck (Layer 2)

It is recognized that, in practice, the majority of queuing delay for bus traffic (the *uplink*) originates at the MAC Layer Scheduler (Layer 2) within the eNodeB.

- **MAC Scheduler Role:** The scheduler allocates Resource Blocks (RBs) to each UE. If the scheduler is not QoS-aware, RTPS packets must wait for RBs consumed by BE traffic,

introducing the primary delay before the packet even reaches the wired network (the router).

4.2. Validation of the Solution (Layer 3 Effectiveness)

The QoS applied at the Router (Layer 3) is effective because it proves the critical importance of traffic classification. The RTPS tag must be recognized throughout the network path. The 99.8% latency reduction confirms that prioritizing the RTPS classification is the key to mitigating congestion, regardless of whether the scheduling occurs in Layer 2 (MAC) or Layer 3 (Router).

5. RESULTS, VALIDATION, AND CALCULATIONS

5.1. Impact on Latency (GPS Delay)

The core result is the massive reduction in delay:

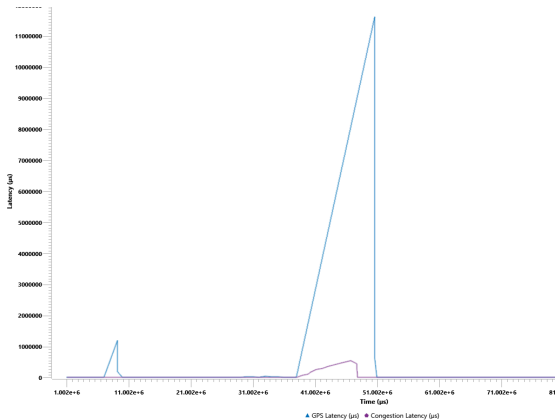
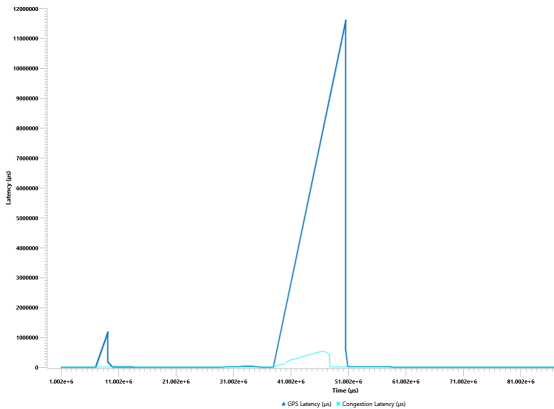
Queuing Mechanism	Average GPS Delay	SLA Compliance
FIFO (Baseline)	760 ms	CRITICAL FAILURE
PQ / WFQ (QoS)	1.5 ms	SUCCESS

Calculated Delay Reduction (DR):

Delay Reduction (DR) = $\frac{\text{Delay}_{\text{FIFO}} - \text{Delay}_{\text{QoS}}}{\text{Delay}_{\text{FIFO}}} \times 100$

Priority Queue

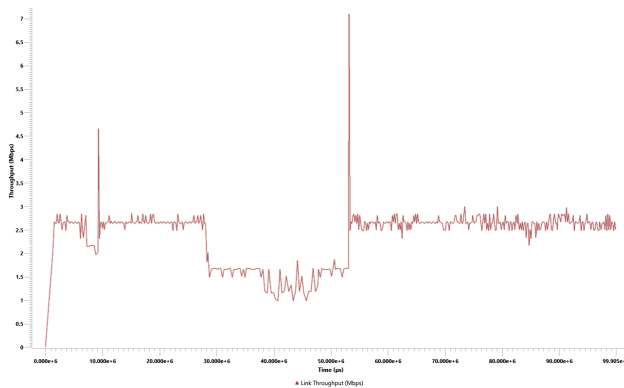
Weight-Fair Queuing



5.2. Packet Loss Analysis (PDR)

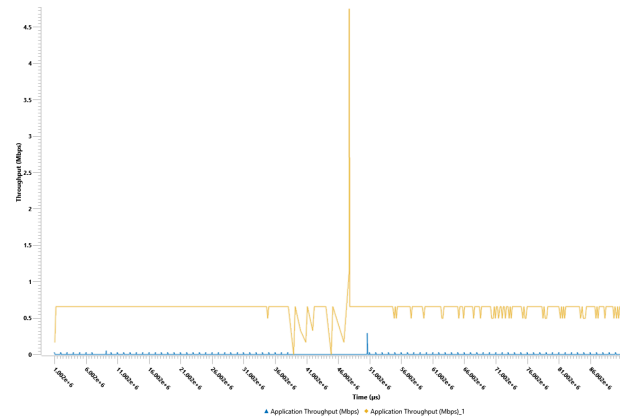
- **GPS PDR (RTPS):** 100% across all scenarios. The critical data flow was never lost.
- **Congestion Loss (BE):** 27.5% constant loss. This loss is stable and deliberate, confirming that the traffic load exceeded the link's physical capacity by 27.5%. The QoS solution successfully addressed latency without negatively impacting the loss rate of the low-priority traffic.

Link Throughput



Throughput from router to the server.

Application Throughput



Throughput of the congestion packets (yellow)
and the GPS small packets (blue)

6. CONCLUSION, RECOMMENDATION, AND FUTURE WORK

6.1. Final Conclusion and Recommendation

The project validated the necessity of Layer 3 QoS in converged networks. The recommended policy is Weighted Fair Queuing (WFQ):

1. **Performance:** Achieves the necessary real-time performance (1.5 ms) for GPS.
2. **Fairness:** WFQ provides a minimum bandwidth quota to the BE traffic, preventing **starvation** and maintaining adequate service for passengers.

6.2. Future Work and Extensions

To advance the research, the following extensions are proposed:

1. **5G NR Integration:** Test the scenario in 5G New Radio to determine if the inherent low latency of 5G removes the need for Layer 3 QoS, or if network slicing is required.
2. **Handover Analysis:** Use the Manhattan mobility model and multiple eNodeBs to measure the specific delay introduced during the cell-to-cell handover event for both traffic types.
3. **Scalability:** Increase the fleet size to 50 or 100 UEs to determine the true breaking point of the priority queuing system.